



Cloud Mushroom Operations Centre

3C Master Controller — User Manual

Firmware v9 | ESP32-S3 Platform

System: IoT Mushroom Cultivation Automation

Hardware: ESP32-S3 DevKitC + ESP32 DOIT V1

Comms: ESP-NOW + WiFi + Telegram Bot

Sensors: SHT40 (T/H) + MH-Z19E (CO2)

Table of Contents

1.	System Overview
2.	Hardware Architecture
3.	Getting Started
4.	Physical Controls — Buttons
5.	LCD Display Screens
6.	NeoPixel Status Indicator
7.	Status LEDs
8.	Telegram Bot — Remote Control
9.	Automation Logic
10.	Threshold Configuration
11.	Alarm & Alert Conditions
12.	Troubleshooting
13.	Technical Specifications

1 System Overview

The **Cloud Mushroom Operations Centre** is an IoT-based environmental automation system designed for commercial mushroom cultivation. It continuously monitors temperature, humidity, and CO2 levels inside the grow room and automatically controls a humidifier and exhaust fan to maintain ideal growing conditions. The system also supports full remote monitoring and manual override via a Telegram bot, and provides real-time feedback through an LCD display, status LEDs, and a NeoPixel RGB indicator.

System Architecture

3C Master Node	2B Sensor Node
ESP32-S3 DevKitC • WiFi + ESP-NOW • 16×2 LCD display • 3× buttons (SET mode) • NeoPixel RGB indicator • 3× status LEDs • Telegram bot • Exhaust fan relay (GPIO4) • Humidifier SSR (GPIO5)	ESP32 DOIT DevKit V1 • ESP-NOW slave • SHT40 — Temperature + Humidity • MH-Z19E — CO2 (UART) • Humidifier SSR (GPIO12) • Exhaust fan relay (GPIO14) • Sends SensorPacket every 10s

***Note:** 3C and 2B communicate wirelessly via ESP-NOW on WiFi channel 11. 3C connects to your WiFi router first, locks the channel, then 2B follows.*

2 Hardware Architecture

3C Master Node — Pin Map

GPIO	Direction	Function
GPIO4	Output	Exhaust fan relay (active HIGH)
GPIO5	Output	Humidifier SSR (active HIGH)
GPIO6	Input	Blue button — NEXT / Enter SET / Confirm save
GPIO7	Input	Red button — Value UP (hold for fast repeat)
GPIO15	Input	Black button — Value DOWN / Cancel SET
GPIO8	I2C SDA	LCD 0x27 + DS3231 RTC + AT24C32 EEPROM
GPIO9	I2C SCL	LCD 0x27 + DS3231 RTC + AT24C32 EEPROM
GPIO12	Output	Red LED — alert / error indicator
GPIO13	Output	Green LED — healthy / all-OK indicator
GPIO20	Output	Yellow LED — MANUAL mode indicator
GPIO48	Output	NeoPixel WS2812B RGB indicator

2B Sensor Node — Pin Map

GPIO	Direction	Function
GPIO21	I2C SDA	SHT40 temperature + humidity sensor
GPIO22	I2C SCL	SHT40 temperature + humidity sensor
GPIO16	UART2 RX	MH-Z19E CO2 sensor
GPIO17	UART2 TX	MH-Z19E CO2 sensor
GPIO12	Output	SSR — humidifier (active HIGH)
GPIO14	Output	Mechanical relay — exhaust fan (active HIGH)

ESP-NOW Communication

- Protocol: ESP-NOW (IEEE 802.11 peer-to-peer, no router hop)
- Channel: 11 (locked to WiFi router 'Ram')
- 3C MAC: 1C:DB:D4:46:22:88 | 2B MAC: D0:EF:76:33:69:48
- 2B sends a 21-byte SensorPacket every 10 seconds and on-demand
- 3C sends CmdPacket structs to 2B to control relay outputs
- Link timeout: if no packet received in 15 seconds, link is marked LOST

3 Getting Started

First Power-On Sequence

Step	What Happens	Expected Indicator
1	3C powers on	NeoPixel: Magenta steady — Booting
2	Connecting to WiFi	NeoPixel: Orange blink
3	WiFi connected	NeoPixel: Green steady (1 second)
4	ESP-NOW initialised	Serial: 'Peer 2B registered'
5	Channel announced to 2B	Serial: 'Channel announced: 11'
6	Waiting for first 2B packet	NeoPixel: Blue steady — no 2B data yet
7	2B packet received	NeoPixel: Bright green pulse, then idle colour
8	System running normally	NeoPixel: Dim green (AUTO) or dim cyan (MANUAL)

Telegram Bot First Use

- Open Telegram and search for your bot (or use the saved link from setup)
- Send /start — you will receive a welcome message confirming 3C is online
- Send /status to verify live sensor readings are flowing
- Send /help to see the full command list at any time

Note: 3C sends a startup notification automatically every time it boots: '3C v9 online SSID: Ram'. If you do not receive this, check WiFi credentials.

4 Physical Controls — Buttons

3C has three physical push-buttons for on-device threshold configuration. Button presses are only active outside of normal operation — they do not interfere with automation, Telegram, or ESP-NOW.

Button Map

Button	Colour	GPIO	Normal Mode	SET Mode
Blue	Blue	6	Single press → Enter SET mode	Advance to next parameter / Save and exit on 3rd press
Red	Red	7	No action	Value +1 (hold for fast repeat after 600ms)
Black	Black	15	No action	Value -1 (hold for fast repeat after 600ms) / Press at first param = Cancel

SET Mode Navigation Flow

Press Blue	→ Enter SET mode. Screen shows Hum Upper %
Red / Black	→ Adjust value up or down
Press Blue	→ Move to Hum Lower %
Red / Black	→ Adjust value
Press Blue	→ Move to CO2 Limit ppm
Red / Black	→ Adjust value
Press Blue	→ SAVE all values and exit SET mode
Press Black (at first param)	→ CANCEL — discard all changes
10s inactivity	→ Auto-save and exit SET mode

SET Mode Value Limits

Parameter	Minimum	Maximum	Step	Notes
Hum Upper %	51 %	99 %	1 %	Must always be > Hum Lower
Hum Lower %	30 %	95 %	1 %	Must always be < Hum Upper
CO2 Limit ppm	400 ppm	5000 ppm	50 ppm	400 ppm is ambient outdoor CO2

5 LCD Display Screens

The 16×2 I2C LCD cycles automatically through three information screens every 3 seconds. In SET mode the LCD is taken over to show the parameter being edited.

Normal Mode — 3 Rotating Screens

Screen	Row 1	Row 2
Screen 1 — Sensors	T: 24.5C H: 82%	CO2: 850 ppm
Screen 2 — Relay & Mode	Fan:OFF Hum:ON	Mode: AUTO
Screen 3 — Link & Thresholds	2B:LIVE Up: 342s	HH:85 HL:75 C:1200

SET Mode Screens

SET Screen	Row 1	Row 2 (example)
Hum Upper %	Hum Upper %	[85.0] BI=Nxt
Hum Lower %	Hum Lower %	[75.0] BI=Nxt
CO2 Limit ppm	CO2 Limit ppm	[1200] BI=Nxt
Saved	Saved!	(blank)
Cancelled	Cancelled	(blank)

Note: 'BI=Nxt' on row 2 is a reminder that Blue button advances to the next parameter.

6 NeoPixel Status Indicator

A single WS2812B NeoPixel (GPIO48) provides at-a-glance system status through 18 distinct colour states, including steady colours, blinks, pulses, and flash sequences.

Complete Colour Reference

Colour	State	Condition / Meaning
	Magenta — steady	System is booting
	Orange — blinking	Connecting to WiFi
	Green — steady (1s)	WiFi just connected
	Red — blinking	WiFi connection lost
	Blue — steady	ESP-NOW active, no 2B data received yet
	Dim green — steady	All OK — system running in AUTO mode
	Dim cyan — steady	All OK — system running in MANUAL mode
	Yellow — steady	CO2 level above set limit
	Light blue — steady	Humidity below lower threshold
	Purple — steady	Humidity above upper threshold
	Orange — steady	CO2 AND humidity alert at the same time
	Red — steady	2B sensor node link lost (15s timeout)
	Bright green — pulse	Data packet received from 2B
	Blue-white — pulse	Command sent to 2B relay
	Cyan — steady	Inside SET mode (button configuration)
	White — single flash	Value changed inside SET mode
	Green — 2 flashes	Thresholds saved successfully
	Red — 2 flashes	SET mode cancelled — changes discarded

Note: Pulse states (bright green, blue-white) last approximately 350ms then return to the appropriate idle colour. Flash states (white, green x2, red x2) complete their sequence then return to idle. SET mode (cyan steady) locks the pixel until you exit.

7 Status LEDs

Three discrete LEDs on the 3C board provide a simple at-a-glance health summary independent of the NeoPixel.

LED	GPIO	Colour	ON when...	OFF when...
LED 1	13	Green	2B link is alive AND no active alerts	Link lost or any alert active
LED 2	12	Red	Any alert: link lost / CO2 high / humidity out of range	All conditions within limits
LED 3	20	Yellow	System is in MANUAL mode	System is in AUTO mode

LED State Summary Table

Scenario	Green	Red	Yellow
2B link OK, all values in range, AUTO	ON	OFF	OFF
2B link OK, all values in range, MANUAL	ON	OFF	ON
CO2 above limit	OFF	ON	OFF or ON
Humidity out of range	OFF	ON	OFF or ON
2B link lost	OFF	ON	OFF or ON
No 2B data received yet	OFF	OFF	OFF or ON

8 Telegram Bot — Remote Control

The Telegram bot allows full remote monitoring and control of the system from any smartphone or computer with Telegram installed. All commands are plain text — no app installation is required beyond Telegram itself.

Setup

- Open Telegram and locate the bot for your system (provided during installation)
- Send /start to register your session and receive a welcome message
- 3C will also send an automatic message each time it reboots

Complete Command Reference

Information Commands

Command	Description
/start	Sends a welcome message and confirms 3C is online
/help	Displays the full command list
/status	Live readings: temperature, humidity, CO2, relay states, mode, 2B link
/thresholds	Shows the currently saved Hum High, Hum Low, and CO2 limit values
/uptime	Shows how long 3C and 2B have been running since last reboot
/link	2B link status, last packet age, error flags, CO2 warm-up state
/rgb	Prints the NeoPixel colour guide directly in the chat

Relay Control Commands

Note: Sending any relay command automatically switches the system to MANUAL mode. Use /auto to hand control back to the automation engine.

Command	Action
/fan on	Turn exhaust fan ON (switches to MANUAL mode)
/fan off	Turn exhaust fan OFF (switches to MANUAL mode)
/hum on	Turn humidifier ON (switches to MANUAL mode)
/hum off	Turn humidifier OFF (switches to MANUAL mode)
/all on	Turn both fan AND humidifier ON (500ms stagger between them)
/all off	Turn both fan AND humidifier OFF (500ms stagger between them)
/auto	Switch to AUTO mode — automation engine takes control
/manual	Switch to MANUAL mode — automation engine is paused

Threshold Configuration Commands

These commands update the automation thresholds remotely without needing physical access to the device. New values are saved to non-volatile storage (NVS) immediately and survive reboots.

Command (example)	Effect
/set humhigh 88	Set humidity upper threshold to 88 % (valid range: 51–99, must be > humlow)
/set humlow 72	Set humidity lower threshold to 72 % (valid range: 30–95, must be < humhigh)

/set co2 1000	Set CO2 alert limit to 1000 ppm (valid range: 400–5000 ppm)
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Example /status Response

```
--- Live Status --- Temp : 24.5 C Hum : 81.0 % CO2 : 870 ppm Fan : OFF Hum : ON Mode :  
AUTO 2B : LIVE ERR : 0x00
```

9 Automation Logic

When AUTO mode is active, 3C evaluates sensor readings every 3 seconds and makes relay decisions based on the saved thresholds.

Decision Rules

Sensor Reading	Condition	Action
CO2	co2ppm > CO2 Limit	Exhaust fan turns ON
CO2	co2ppm <= CO2 Limit	Exhaust fan turns OFF
Humidity	humidity < Hum Lower AND humidifier is currently OFF	Humidifier turns ON
Humidity	humidity > Hum Upper AND humidifier is currently ON	Humidifier turns OFF

AUTO vs MANUAL Mode

- In AUTO mode: relays are fully controlled by the above rules. Manual Telegram commands are ignored for relays.
- In MANUAL mode: automation is paused. Relays hold their last commanded state until you send a new command.
- Any relay command via Telegram (/fan on, /hum off, etc.) automatically switches to MANUAL mode.
- Send /auto at any time to re-enable automatic control.
- Mode is not saved across reboots — the system always starts in AUTO mode.

Safety Features

- Relays default to OFF (safe-low) on boot — nothing activates until valid 2B data is received
- Automation only runs if got2BData is true — prevents stale decisions
- 500ms relay stagger on /all commands prevents simultaneous current surges
- 1N4007 flyback diodes fitted across each relay coil to suppress voltage spikes
- AC mains switched only through relay COM/NO screw terminals — never Dupont connectors

10 Threshold Configuration

Thresholds can be changed in two ways: physically using the onboard buttons (SET mode), or remotely via Telegram /set commands. Both methods save values to NVS immediately and values survive power cuts and reboots.

Default Values

Parameter	Default	Recommended Range	Notes
Hum High	85 %	80–92 %	Humidifier turns OFF above this
Hum Low	75 %	70–82 %	Humidifier turns ON below this
CO2 Limit	1200 ppm	800–1500 ppm	Fan turns ON above this limit

Note: The hysteresis gap between Hum Low and Hum High prevents rapid relay cycling. A gap of at least 5–10 % is recommended.

11 Alarm & Alert Conditions

The system monitors for four alert conditions. When any alert is active, the Red LED turns ON and the Green LED turns OFF.

Alert	Trigger Condition	NeoPixel Colour	Relay Response
CO2 High	CO2 reading > CO2 Limit	Yellow steady	Fan turns ON (AUTO)
Humidity Low	Humidity < Hum Low threshold	Light blue steady	Humidifier turns ON (AUTO)
Humidity High	Humidity > Hum High threshold	Purple steady	Humidifier turns OFF (AUTO)
2B Link Lost	No packet for > 15 seconds	Red steady	No change — last state held
CO2 + Humidity	CO2 high AND humidity out of range	Orange steady	Both actions apply (AUTO)

Telegram Alerts

- Currently the system does not send proactive Telegram alerts — monitoring is pull-based (send /status to check)
- Use /status regularly or set a reminder to check during the cultivation cycle
- The /link command is the fastest way to check if the sensor node is alive

12 Troubleshooting

Symptom	Likely Cause	Resolution
NeoPixel stays blue after boot	2B has not sent any data yet	Check 2B power. Check ESP-NOW channel both nodes on ch.11. Check 2B firmware running.
NeoPixel red blink	WiFi connection lost	Check router power and SSID. 3C retries every 30s automatically.
NeoPixel red steady	2B link timeout (>15s no packet)	Check 2B power. Check physical distance and obstructions.
No Telegram response	3C not connected to WiFi	Check NeoPixel — red blink = no WiFi. Check router.
/status shows 0.0 for all values	2B data not yet received	Wait for first packet or check 2B status.
Humidity reads 0% or unrealistic	SHT40 I2C error on 2B	Check 2B serial output. Verify GPIO21/22 wiring on 2B.
CO2 reads 0 ppm	MH-Z19E still in warm-up (3–5 min)	Wait for preheat to complete. /link shows 'CO2 warm: YES' during this.
Relay not responding to Telegram	Command sets MANUAL but 2B not acting	2B receives CMD over ESP-NOW — check 2B serial output for CMD log.
SET mode saves wrong values	10s timeout auto-saved before done	Work faster or note: hold Red/Black for fast repeat to reach target quickly.

13 Technical Specifications

Firmware

Item	Value
Firmware version	3C v9 / 2B stable
Platform	ESP32 Arduino Core 3.3.x
Board (3C)	ESP32-S3 DevKitC — Xtensa LX7 dual-core 240MHz
Board (2B)	ESP32 DOIT DevKit V1 — Xtensa LX6 dual-core 240MHz
Flash	4 MB (both nodes)
Upload speed	921600 baud
Threshold storage	NVS (non-volatile) — survives reboots and power cuts

Sensors

Sensor	Parameter	Range	Accuracy	Interface
SHT40	Temperature	-40 to +125 C	+/- 0.2 C	I2C (0x44)
SHT40	Humidity	0–100 % RH	+/- 1.8 %	I2C (0x44)
MH-Z19E	CO2	0–5000 ppm	+/- 50 ppm	UART 9600

Connectivity

Item	Value
WiFi	2.4 GHz 802.11 b/g/n (SSID: Ram, Channel: 11)
ESP-NOW	802.11 peer-to-peer, channel 11, unencrypted
Telegram polling	Every 3 seconds (long-poll timeout=0)
Heartbeat to 2B	CMD_REQ_DATA every 10 seconds
Channel announce	Every 30 seconds

Libraries Used

Library	Purpose	Source
Adafruit NeoPixel	WS2812B RGB LED control	Arduino Library Manager
LiquidCrystal_I2C	16x2 LCD via I2C	Frank de Brabander
Preferences	NVS read/write	Built-in ESP32 core
WiFiClientSecure	HTTPS for Telegram API	Built-in ESP32 core
HTTPClient	HTTP POST/GET for Telegram	Built-in ESP32 core
esp_now.h	ESP-NOW peer comms	ESP-IDF (via Arduino core)
esp_wifi.h	WiFi channel management	ESP-IDF (via Arduino core)

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